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About

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Scaling AI into the Physical World

Artificial Intelligence (AI) has made tremendous advances in the past several years, driven largely by U.S. innovation. However, **impressive scores on AI benchmark tests alone will not secure strategic competitiveness or build national power.** The Trump Administration should prioritize ensuring the US is a leader in not just developing AI models, but **scaling AI into the real world**, integrating AI systems in key domains relevant for strategic competition.

In this response, we identify three key areas where this administration can focus attention to accelerate the diffusion and adoption of AI systems to support US leadership: next-generation AI chip designs, autonomous power grids, and accelerating real-world scientific discoveries.

Next Generation AI Chip Designs

The CHIPS and Science Act helped onshore advanced semiconductor manufacturing capabilities and bolster a key gap in the American supply chain. However, China continues to advance in its domestic development of semiconductor manufacturing through SMIC and chip design capabilities through Huawei and other companies. Not only does China aim to close the gap in semiconductor manufacturing, it is also seeking to erode U.S. leadership in chip design. Since the first chip export control rules limiting PRC access to Nvidia GPUs, there has been a proliferation of new Chinese AI chip companies and PRC funding to develop new designs of AI accelerators. Today, China is the world leader in high quality and top-cited research on AI chip design¹ and is continuing to use public capital to support new AI chip startups.

The US should take steps to avoid strategic surprise in AI hardware design.

Export controls may, as a second order effect, create conditions for our competitors to invest in and explore different technological paths towards a next-generation AI chip, potentially co-designed with new AI model architectures, and leapfrog US capabilities. A robust “compute industrial base” is needed to avoid strategic surprise, not just in terms of advanced semiconductor manufacturing, but also in terms of a multi-vendor AI chip ecosystem.

The US government can play a crucial role in supporting adoption of AI technologies to accelerate chip design iteration and innovation. This will not only boost US leadership in AI chips, but also help support the development of more energy efficient and performant AI chips, which can reduce the energy requirements and cost for AI training runs.

Finally, there are a number of proposals to impose security requirements onto AI chip designers. Many of these proposals promise that such requirements come at “no cost to the government”, eliding the fact that such requirements would impose enormous costs on chip design companies, hurting US chip competitiveness while also adding surveillance technologies onto these chips that jeopardize global trust in US-led AI development. The Trump administration should carefully balance security concerns with the importance of global competitiveness and be mindful of imposing burdensome regulations that inhibit technointustrial development.

¹“The state of global chip research,” Emerging Technology Observatory, March 3, 2025, <https://www.eto.tech/blog/state-global-chip-research/>.

Key recommendations:

- Support Department of Energy national lab AI testbeds², which serve as a customer-of-first-resort for many early stage AI chip companies, such as Cerebras, Sambanova, and Groq. These testbeds also give our national labs and other US scientists early access to advanced AI hardware to run novel scientific AI models
- Leverage the Department of Defense's varied mission requirements as an early market for leveraging AI to develop domain-specific chips. For instance, DIU³ and DARPA⁴ have released solicitations in the past year to design new high-performance, mission-specific chip chips. DARPA could replicate their self-driving car Grand Challenge⁵ to create a competition for AI designed chips that meet certain mission requirements.

²"Artificial Intelligence Testbeds at DOE," U.S. Department of Energy,
<https://www.energy.gov/cet/artificial-intelligence-testbeds-doe>.

³Cal Biesecker, "DIU Seeks Digital Platform to Help Speed Design and Validation of Analog Chips," *Defense Daily*, February 10, 2025,
<https://www.defensedaily.com/diu-seeks-digital-platform-to-help-speed-design-and-validation-of-analog-chips/advanced-transformational-technology/>.

⁴Sydney J. Freedberg Jr., "DARPA's OPTIMA Program Seeks Ultra-Efficient AI Chips," *Breaking Defense*, March 20, 2024,
<https://breakingdefense.com/2024/03/darpas-optima-program-seeks-ultra-efficient-ai-chips/>.

⁵"The DARPA Grand Challenge: Ten Years Later," *DARPA News*, March 13, 2014,
<https://www.darpa.mil/news/2014/grand-challenge-ten-years-later>.

Autonomous Power Grids

Building frontier scale AI models in the US increasingly requires the rapid deployment of domestic AI training data centers, which often require 1GW+ of power. While the US benefits from deep capital markets that are willing to invest in data centers and the powerful AI chips in them, the US power market has been sluggish to respond to the addition of large-scale power loads. While CIS has written about steps the Trump Administration can take to scale Federal Hydropower to realize Energy Dominance and support load growth⁶, AI can also play a role in helping meet the very load growth it creates.

A growing grid is key to a growing economy. Part of China's success in industrialization is their dynamic power grid, which has grown at over 6% every year⁷, compared to the U.S. which is struggling to maintain a new pace of 2% in year over year growth⁸ in power demand over the past several recent years. Power is critical not only for data centers, but also smelters, refineries, and manufacturers critical to building U.S. industrial dominance.

Key bottlenecks in adding new power generation include federal permitting⁹ and a sluggish interconnection process for connecting new power¹⁰. **AI can help accelerate every part of grid processes**, down to the physics models used to operate the grid. New utility industry alliances¹¹ are already beginning to form to support adoption of new grid technologies that incorporate AI. While AI data centers are driving load growth, AI can also help solve the very problems it creates by creating a modern, autonomous power grid.

⁶Charles Yang, "Trump's National Energy Dominance Council Should Appoint a 'Hydropower Czar'," *Utility Dive*, February 25, 2025,

<https://www.utilitydive.com/news/trump-national-energy-dominance-council-hydropower/740801/>.

⁷Lin Boqiang, "New developments in the recent low-carbon transition of electricity in China," *World Economic Forum*, January 15, 2025,

<https://www.weforum.org/stories/2025/01/low-carbon-transition-of-energy-electricity-china/>.

⁸"EIA publishes its first energy-sector forecasts through 2026", Energy Information Administration, January 14, 2025. <https://www.eia.gov/pressroom/releases/press564.php>

⁹Thomas Hochman, "The Energy Permitting Reform Act Doesn't Go Far Enough," *Foundation for American Innovation*, October 29, 2024,

<https://www.thefai.org/posts/the-energy-permitting-reform-act-doesn-t-go-far-enough>.

¹⁰Dan McCarthy and Maria Virginia Olano, "Chart: The US Clean Energy Backlog Is Getting Bigger and Bigger," *Canary Media*, April 19, 2024,

<https://www.canarymedia.com/articles/transmission/chart-the-us-clean-energy-backlog-is-getting-bigger-and-bigger>.

¹¹"Orchestrating the Grid to Deliver the Energy Transition," *TGO Alliance*, <https://tgoalliance.org/>.

Key recommendations:

- The DOE can leverage remaining funding through the Grid Deployment Office to support commercial deployment of AI to accelerate grid interconnection processes.¹² Interconnection queue timelines are one of the primary barriers to more rapidly deploying all-of-the-above energy generation; DOE can help derisk adoption of new AI tools to address this key bottleneck.
- The Council for Environmental Quality (CEQ) can accelerate the deployment of AI for permitting tools, like Pacific Northwest National Lab's PolicyAI¹³, across federal regulatory agencies. These tools help augment the environmental permitting workforce and can help streamline approvals for permits.

¹²"AI for Interconnection (AI4IX)," *U.S. Department of Energy*, November 25, 2024, <https://www.energy.gov/gdo/ai-interconnection-ai4ix>.

¹³"PolicyAI," *Pacific Northwest National Laboratory*, <https://www.pnnl.gov/projects/policyai>.

Bringing AI into the Lab

AI will be a key accelerant for scientific discovery. However, current models today are unable to realize or actuate in the physical world. **In a world where inference and ideas are cheap, testing and verification become the bottleneck for scientific progress.** The bottleneck to discovering new semiconductor materials, new cancer drugs, and new high-performance ceramics for hypersonics in the age of AI is no longer ideas, but rather the ability for scientists to test and verify predictions in the real world.

Autonomous experimentation infrastructure is the linchpin to bring AI into the lab. Combining robotics, automation, and AI-driven predictive models, autonomous experimentation enables closed-loop cycles of hypothesis testing, synthesis, and characterization of new materials. Tech companies are already beginning to recognize the importance of so-called “[self-driving labs](#)” in advancing AI for science: [Meta worked with University of Toronto](#) to autonomously test hundreds of catalyst designs through their self-driving labs, after they had already developed an AI materials model trained on simulation data. However, autonomous experimentation is high capital intensity and still nascent - public R&D investment will be required to scale this linchpin technology to realize AI-accelerated science.

Key recommendations:

- The Office of Science and Technology Policy (OSTP) can coordinate through the interagency Material Genome Institute (MGI) to create a consortium for self-driving labs, bringing together scientific equipment providers, robotics manufacturers, scientists, and AI developers, to develop a roadmap for “Grand Challenges in Autonomous Scientific Discovery”. MGI can leverage responses to the recent [Request for Information the Department of Energy released on autonomous experimentation](#) to develop this roadmap.
- The Department of Energy (DOE) Office of Science should increase funding support for national labs to invest in autonomous experimentation infrastructure for self-driving labs. The Trump administration should also work with Congress to pass the bipartisan DOE AI Act, which would support the development of a national AI capability through the DOE and its national labs. DOE and its 17 national labs are well-positioned to integrate and scale AI through scientific infrastructure and its national security mission.